

		LLM	CodeAct	RLM	LLM	CodeAct	RLM	CodeAct	RLM	RLM	
		100	100	100	500	500	500	1000	1000	Full	
T1	T1.1 Exact product lookup (10Q)	0.92	1.00	0.95	0.54	1.00	0.91	1.00	0.76	0.84	Macro F1
	T2.1 Molecular-weight change (6Q)	0.07	0.88	0.83	0.02	0.84	0.80	0.69	0.57	0.87	
	T2.2 Ring-count change (5Q)	0.10	0.92	0.92	0.16	0.92	0.84	0.79	0.68	0.84	
	T2.3 Aromatic-ring formation (5Q)	0.04	0.96	0.89	0.01	0.92	0.92	0.76	0.82	0.84	
	T2.4 Combined weight + rings (4Q)	0.02	0.90	0.85	0.04	0.75	1.00	0.25	0.75	0.81	
T3	T3.1 New nitrogen atom (1Q)	0.13	1.00	0.80	0.14	1.00	1.00	0.82	1.00	0.80	
	T3.2 C-N formed / C-O broken (1Q)	0.31	1.00	0.60	0.31	1.00	0.60	0.00	0.60	0.80	
	T3.3 Single C-C bond formation (1Q)	0.00	0.82	0.40	0.00	0.63	0.80	0.15	0.60	0.80	
	T3.4 Fused heterocycle (1Q)	0.08	0.77	0.33	0.01	0.80	0.40	0.00	0.28	0.68	
	T3.5 New ring system (1Q)	0.54	0.55	0.00	0.00	0.56	0.39	0.47	0.33	0.22	
T3	T3.6 Fused-ring construction (1Q)	0.05	1.00	0.40	0.07	0.68	1.00	0.60	0.80	1.00	
	T3.7 Transition-metal reagent (1Q)	0.61	1.00	0.60	0.34	0.94	0.80	0.00	0.80	0.60	
	T3.8 HATU / T3P reagent (1Q)	0.96	0.91	0.82	0.00	0.98	1.00	0.77	0.60	0.20	
	T3.9 New stereocenter (1Q)	0.10	1.00	0.80	0.02	1.00	1.00	0.00	1.00	0.80	
	T3.10 E-alkene formation (1Q)	0.06	1.00	0.60	0.06	0.98	0.40	0.80	0.80	0.37	
T3	T3.11-14 Amide couplings (4Q)	0.01	0.82	0.68	0.04	0.54	0.53	0.00	0.20	0.47	
	T3.15-19 Group transformations (5Q)	0.73	0.93	0.79	0.34	0.69	0.91	0.10	0.41	0.73	
	T3.20-21 Protecting groups (2Q)	0.45	0.94	0.77	0.01	0.85	0.44	0.00	0.40	0.66	
	T3.22-25 Named reactions (4Q)	0.33	0.64	0.88	0.04	0.53	0.93	0.12	0.27	0.46	
	T3.26-30 Mechanisms I (5Q)	0.25	0.58	0.82	0.21	0.19	0.82	0.10	0.37	0.37	
T3	T3.31-35 Mechanisms II (5Q)	0.39	0.78	0.63	0.28	0.65	0.27	0.33	0.22	0.11	
	T4.1-2 Fixed-length chains (2Q)	0.07	0.66	0.73	0.05	0.78	0.80	0.77	0.60	0.70	
	T4.3-4 Longest chains (2Q)	0.38	0.84	0.60	0.05	0.65	0.70	0.92	0.80	0.70	
	T4.5 Hub molecules (1Q)	0.13	0.96	0.80	0.00	0.60	0.40	0.88	1.00	0.80	
	T4.6-9 Group-constrained chains (4Q)	0.25	0.70	0.55	0.05	0.85	0.30	0.35	0.60	0.39	
T4	T4.10-11 Protecting-group pairs (2Q)	0.00	0.37	0.40	0.00	0.27	0.34	0.40	0.40	0.09	
	T4.12-15 Ring-construction chains (4Q)	0.00	0.52	0.00	0.00	0.42	0.20	0.00	0.70	0.45	
	T4.16-25 Truncated routes (10Q)	0.14	0.00	0.45	0.05	0.00	0.32	0.00	0.30	0.04	
	T4.26-30 SMIRKS chains I (5Q)	0.08	0.12	0.73	0.00	0.00	0.60	0.00	0.59	0.00	
	T4.31-35 SMIRKS chains II (5Q)	0.24	0.21	0.34	0.03	0.28	0.25	0.09	0.57	0.00	